Georgii Melidi

Data Scientist | Applied Machine Learning and Optimization
Paris, France | georgii.melidi@gmail.com | gmelidi.com
github.com/georgiimelidi | Google Scholar

Profile

I hold a PhD in Computer Science from Sorbonne Université and combine mathematical modelling, Python development, and applied machine learning. My experience includes banking transaction classification, industrial optimization, and independent AI applications. I enjoy turning complex questions into working code, testing assumptions, and explaining results clearly. Running a customer-facing business has also given me practical experience in pricing, budgets, and day-to-day operations.

Technical Skills

Programming and data: Python, SQL, C++; pandas, NumPy, SciPy, Matplotlib; data preprocessing and reusable computational pipelines.
Quantitative methods: robust and combinatorial optimization, MILP with Gurobi and CPLEX, probability, statistics, simulation, sensitivity analysis, and baseline comparisons.
ML and applied AI: scikit-learn, PyTorch, Hugging Face; classification, feature engineering, embeddings, retrieval, and contextual analysis agents.
Development: Git/GitHub, Docker, Streamlit; reproducible experiments, modular pipelines, interactive applications, and technical documentation.

Research and Applied Experience

Sorbonne Université / CNRS LIP6, Paris 2022 - 2025
PhD Researcher, Operations Research

- Built modular Python pipelines for scenario generation, simulation, baseline comparisons, and analysis of computational experiments.
- Developed and analysed algorithms using imperfect predictions, examining how performance changed with prediction errors and modelling assumptions.
- Investigated failure cases, documented experimental methods, and worked with collaborators to make results reproducible.
- Collaborated from problem formulation through implementation and evaluation; documented methods and communicated results in papers and presentations.

Gazprom Neft Science and Technology Centre 2020 - 2022
Algorithm Development -- alongside Master's studies

- Developed optimization heuristics for gas processing and transport networks subject to operational constraints.

Center of Financial Technologies (CFT), Novosibirsk 2019 - 2020
Data Science Intern

- Worked on predicting a structured banking-transaction field from transaction records, using Python and NLP for preprocessing, feature extraction, classification, and evaluation.

Level Up Computer Club, Novosibirsk 2019 - 2025
Co-founder and Managing Director

- Managed pricing, budgets, capacity, and day-to-day operations; worked directly with customers and staff to understand service needs and resolve operational problems.

Independent Data Science Projects

RL Dynamic Pricing GitHub
Built a stochastic pricing simulator and trained a PyTorch DQN. Compared learned policies with fixed-price and heuristic baselines under changing demand, examining revenue and robustness.

CourtSense AI GitHub
Built an interactive sports prediction and tournament-simulation application combining Elo and ranking features, retrieval over similar cases, and contextual analysis agents. Used Streamlit to make predictions and alternative scenarios accessible through a working interface.

Education

PhD, Computer Science - Sorbonne Université / LIP6 2022 - 2025
Research: learning-augmented algorithms, robust optimization, and decision-theoretic evaluation.
MSc and BSc, Applied Mathematics and Informatics 2016 - 2022
Novosibirsk State University (NSU).

Selected Publications

ICLR 2026: Decision-Theoretic Approaches for Improved Learning-Augmented Algorithms. Co-author.
AAAI 2025: Scenario-Based Robust Optimization of Tree Structures. Co-author.

Additional Experience

Teads Data Science Summer School, 2026 (participant): applied ML, experimentation, and model evaluation.
Languages: English (fluent), French (intermediate), Russian (native).